

CodeGuard Dataset - Source Code

Programs 1 to 100 (C: 50, C++: 50). Each program and each line is numbered.

Program 1 student_001.c | C

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     char word[] = "madam";
6     int lo = 0, hi = strlen(word) - 1, ok = 1;
7     while (lo < hi) {
8         if (word[lo] != word[hi]) {
9             ok = 0;
10            break;
11        }
12        lo++;
13        hi--;
14    }
15    printf("%s is %sa palindrome\n", word, ok ? "" : "not ");
16    return 0;
17 }
```

Program 2 student_002.c | C

```
1 #include <stdio.h>
2
3 int main() {
4     int data[6] = {64, 25, 12, 22, 11, 90};
5     int size = 6, holder;
6     for (int p = 0; p < size - 1; p++) {
7         for (int q = 0; q < size - p - 1; q++) {
8             if (data[q] > data[q + 1]) {
9                 holder = data[q];
10                data[q] = data[q + 1];
11                data[q + 1] = holder;
12            }
13        }
14    }
15    printf("Sorted: ");
16    for (int p = 0; p < size; p++) printf("%d ", data[p]);
17    printf("\n");
18    return 0;
19 }
```

Program 3 student_003.c | C

```
1 #include <stdio.h>
2
3 int findHcf(int x, int y) {
4     while (y != 0) {
5         int tmp = y;
6         y = x % y;
7         x = tmp;
8     }
9     return x;
10 }
11
12 int main() {
13     printf("GCD of 48 and 18 = %d\n", findHcf(48, 18));
14     return 0;
15 }
```

Program 4 student_004.cpp | C++

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Account {
6     string owner;
7     double balance;
8 public:
9     Account(string name, double initial) : owner(name), balance(initial) {}
10    void deposit(double amt) { balance += amt; }
11    bool withdraw(double amt) {
12        if (amt > balance) return false;
13        balance -= amt;
14        return true;
15    }
16    void show() { cout << owner << ": " << balance << endl; }
17 };
18
19 int main() {
20     Account acc("Rahul", 1000);
21     acc.deposit(500);
22     if (!acc.withdraw(2000)) cout << "Insufficient funds" << endl;
23     acc.withdraw(300);
24     acc.show();
25     return 0;
26 }
```

Program 5 student_005.c | C

```
1 // Assignment submission by Student 63
2 // Date: 12-09-2026
3
4 #include <stdio.h>
5 int gcd(int a, int b) {
6     while (b != 0) {
7         int t = b; // step 3
8         b = a % b;
9         a = t;
10    }
11    return a; // step 6
12 }
13
14 /* program entry point */
15 int main() {
16     printf("GCD of 48 and 18 = %d\n", gcd(48, 18));
17     return 0; // step 9
18 }
```

Program 6 **student_006.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  class Figure {
5  public:
6      virtual double getArea() = 0;
7      virtual ~Figure() {}
8  };
9
10 class Round : public Figure {
11     double rad;
12 public:
13     Round(double rr) : rad(rr) {}
14     double getArea() { return 3.14159 * rad * rad; }
15 };
16
17 class Box : public Figure {
18     double ln, wd;
19 public:
20     Box(double ba, double bb) : ln(ba), wd(bb) {}
21     double getArea() { return ln * wd; }
22 };
23
24 int main() {
25     Figure *figs[2] = {new Round(5), new Box(4, 6)};
26     for (int k = 0; k < 2; k++) {
27         cout << "Area: " << figs[k]->getArea() << endl;
28         delete figs[k];
29     }
30     return 0;
31 }
```

Program 7 **student_007.cpp | C++**

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  class Account {
6      string owner;
7      double balance;
8  public:
9      Account(string name, double initial) : owner(name), balance(initial) {}
10     void deposit(double amt) { balance += amt; }
11     bool withdraw(double amt) {
12         if (amt > balance) return false;
13         balance -= amt;
14         return true;
15     }
16     void show() { cout << owner << ": " << balance << endl; }
17 };
18
19 int main() {
20     Account acc("Rahul", 1000);
21     acc.deposit(500);
22     if (!acc.withdraw(2000)) cout << "Insufficient funds" << endl;
23     acc.withdraw(300);
24     acc.show();
25     return 0;
26 }
```

Program 8 **student_008.c | C**

```
1 // Assignment submission by Student 28
2 // Date: 12-09-2026
3
4 #include <stdio.h>
5 /* program entry point */
6 int main() {
7     int num = 12345, rev = 0, digit;
8     int temp = num; // step 3
9     while (temp != 0) {
10         digit = temp % 10;
11         rev = rev * 10 + digit; // step 6
12         temp /= 10;
13     }
14     printf("Reverse of %d is %d\n", num, rev);
15     return 0; // step 9
16 }
```

Program 9 **student_009.cpp | C++**

```
1 // Assignment submission by Student 84
2 // Date: 12-09-2026
3
4 #include <iostream>
5 #include <string>
6 using namespace std;
7 class Account {
8     string owner; // step 3
9     double balance;
10 public:
11     Account(string name, double initial) : owner(name), balance(initial) {}
12     void deposit(double amt) { balance += amt; }
13     bool withdraw(double amt) {
14         if (amt > balance) return false; // step 6
15         balance -= amt;
16         return true;
17     }
18     void show() { cout << owner << ": " << balance << endl; }
19 }; // step 9
20 /* program entry point */
21 int main() {
22     Account acc("Rahul", 1000);
23     acc.deposit(500); // step 12
24     if (!acc.withdraw(2000)) cout << "Insufficient funds" << endl;
25     acc.withdraw(300);
26     acc.show(); // step 15
27     return 0;
28 }
```

Program 10 **student_010.cpp | C++**

```
1 #include <iostream>
2 #include <string>
3 #include <cctype>
4 using namespace std;
5
6 int main() {
7     string text = "Programming in Cpp";
8     int count = 0;
9     for (int i = 0; i < text.length(); i++) {
10         char ch = tolower(text[i]);
11         if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') count++;
12     }
13     cout << "Vowels: " << count << endl;
14     return 0;
15 }
```

Program 11 **student_011.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int arr[] = {34, 7, 91, 15, 62};
6      int n = 5;
7      int maxVal = arr[0], minVal = arr[0];
8      for (int i = 1; i < n; i++) {
9          if (arr[i] > maxVal) maxVal = arr[i];
10         if (arr[i] < minVal) minVal = arr[i];
11     }
12     cout << "Max: " << maxVal << " Min: " << minVal << endl;
13     return 0;
14 }
```

Program 12 **student_012.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  template <typename Type>
5  void exchange(Type &first, Type &second) {
6      Type hold = first;
7      first = second;
8      second = hold;
9  }
10
11 int main() {
12     int i1 = 10, i2 = 20;
13     exchange(i1, i2);
14     cout << "x=" << i1 << " y=" << i2 << endl;
15     double d1 = 1.5, d2 = 2.5;
16     exchange(d1, d2);
17     cout << "p=" << d1 << " q=" << d2 << endl;
18     return 0;
19 }
```

Program 13 **student_013.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  class Stack {
5      int arr[100];
6      int top;
7  public:
8      Stack() { top = -1; }
9      void push(int x) { arr[++top] = x; }
10     int pop() { return arr[top--]; }
11     bool isEmpty() { return top == -1; }
12 };
13
14 int main() {
15     Stack s;
16     for (int i = 1; i <= 5; i++) s.push(i * 2);
17     while (!s.isEmpty()) cout << s.pop() << " ";
18     cout << endl;
19     return 0;
20 }
```

Program 14 **student_014.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int terms = 10, first = 0, second = 1, next;
5      printf("Fibonacci series: ");
6      for (int i = 0; i < terms; i++) {
7          printf("%d ", first);
8          next = first + second;
9          first = second;
10         second = next;
11     }
12     printf("\n");
13     return 0;
14 }
```

Program 15 **student_015.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int arr[] = {2, 5, 8, 12, 16, 23, 38, 56};
5      int n = 8, key = 23;
6      int low = 0, high = n - 1, pos = -1;
7      while (low <= high) {
8          int mid = (low + high) / 2;
9          if (arr[mid] == key) {
10             pos = mid;
11             break;
12         } else if (arr[mid] < key) low = mid + 1;
13         else high = mid - 1;
14     }
15     if (pos != -1) printf("Found at index %d\n", pos);
16     else printf("Not found\n");
17     return 0;
18 }
```

Program 16 **student_016.c | C**

```
1  #include <stdio.h>
2
3  unsigned long long factorial(int n) {
4      if (n <= 1) {
5          return 1;
6      }
7      return n * factorial(n - 1);
8  }
9
10 int main() {
11     int num = 10;
12     printf("Factorial of %d = %llu\n", num, factorial(num));
13     return 0;
14 }
```

Program 17 **student_017.c | C**

```
1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char str[] = "madam";
6      int i = 0, j = strlen(str) - 1, isPal = 1;
7      while (i < j) {
8          if (str[i] != str[j]) {
9              isPal = 0;
10             break;
11         }
12         i++;
13         j--;
14     }
15     printf("%s is %sa palindrome\n", str, isPal ? "" : "not ");
16     return 0;
17 }
```

Program 18 **student_018.c | C**

```
1  #include <stdio.h>
2
3  unsigned long long calcFact(int x) {
4      if (x <= 1) {
5          return 1;
6      }
7      return x * calcFact(x - 1);
8  }
9
10 int main() {
11     int value = 10;
12     printf("Factorial of %d = %llu\n", value, calcFact(value));
13     return 0;
14 }
```

Program 19 **student_019.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  struct Node {
5      int data;
6      Node *next;
7  };
8
9  class Queue {
10     Node *front, *rear;
11 public:
12     Queue() { front = rear = nullptr; }
13     void enqueue(int x) {
14         Node *n = new Node{x, nullptr};
15         if (rear == nullptr) front = rear = n;
16         else {
17             rear->next = n;
18             rear = n;
19         }
20     }
21     int dequeue() {
22         Node *t = front;
23         int v = t->data;
24         front = front->next;
25         if (front == nullptr) rear = nullptr;
26         delete t;
27         return v;
28     }
29     bool empty() { return front == nullptr; }
30 };
31
32 int main() {
33     Queue q;
34     for (int i = 1; i <= 5; i++) q.enqueue(i * 3);
35     while (!q.empty()) cout << q.dequeue() << " ";
36     cout << endl;
37     return 0;
38 }
```

Program 20 **student_020.cpp | C++**

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  struct BankAcc {
6      string holder;
7      double money;
8  };
9
10 void credit(BankAcc &b, double x) { b.money += x; }
11
12 bool debit(BankAcc &b, double x) {
13     if (b.money < x) return false;
14     b.money -= x;
15     return true;
16 }
17
18 int main() {
19     BankAcc a = {"Rahul", 1000};
20     credit(a, 500);
21     if (!debit(a, 2000)) cout << "Insufficient funds\n";
22     debit(a, 300);
23     cout << a.holder << ": " << a.money << "\n";
24     return 0;
25 }
```

Program 21 **student_021.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  class Stack {
5      int arr[100];
6      int top;
7  public:
8      Stack() { top = -1; }
9      void push(int x) { arr[++top] = x; }
10     int pop() { return arr[top--]; }
11     bool isEmpty() { return top == -1; }
12 };
13
14 int main() {
15     Stack s;
16     for (int i = 1; i <= 5; i++) s.push(i * 2);
17     while (!s.isEmpty()) cout << s.pop() << " ";
18     cout << endl;
19     return 0;
20 }
```

Program 22 **student_022.c | C**

```
1  #include <stdio.h>
2
3  unsigned long long factorial(int n) {
4      if (n <= 1) {
5          return 1;
6      }
7      return n * factorial(n - 1);
8  }
9
10 int main() {
11     int num = 10;
12     printf("Factorial of %d = %llu\n", num, factorial(num));
13     return 0;
14 }
```

Program 23 **student_023.cpp | C++**

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  class Wallet {
6      string holder;
7      double funds;
8  public:
9      Wallet(string person, double start) : holder(person), funds(start) {}
10     void addMoney(double sum) { funds += sum; }
11     bool takeOut(double sum) {
12         if (sum > funds) return false;
13         funds -= sum;
14         return true;
15     }
16     void print() { cout << holder << ": " << funds << endl; }
17 };
18
19 int main() {
20     Wallet w("Rahul", 1000);
21     w.addMoney(500);
22     if (!w.takeOut(2000)) cout << "Insufficient funds" << endl;
23     w.takeOut(300);
24     w.print();
25     return 0;
26 }
```

Program 24 **student_024.cpp | C++**

```
1 // Assignment submission by Student 140
2 // Date: 12-09-2026
3
4 #include <iostream>
5 using namespace std;
6 struct Node {
7     int data; // step 3
8     Node *next;
9 };
10 class Queue { // step 6
11     Node *front, *rear;
12 public:
13     Queue() { front = rear = nullptr; }
14     void enqueue(int x) {
15         Node *n = new Node{x, nullptr}; // step 9
16         if (rear == nullptr) front = rear = n;
17         else {
18             rear->next = n; // step 12
19             rear = n;
20         }
21     }
22     int dequeue() {
23         Node *t = front; // step 15
24         int v = t->data;
25         front = front->next;
26         if (front == nullptr) rear = nullptr; // step 18
27         delete t;
28         return v;
29     }
30     bool empty() { return front == nullptr; }
31 }; // step 21
32 /* program entry point */
33 int main() {
34     Queue q;
35     for (int i = 1; i <= 5; i++) q.enqueue(i * 3); // step 24
36     while (!q.empty()) cout << q.dequeue() << " ";
37     cout << endl;
38     return 0; // step 27
39 }
```

Program 25 **student_025.cpp | C++**

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  class Student {
6      string name;
7      int marks[3];
8  public:
9      Student(string n, int m1, int m2, int m3) {
10         name = n;
11         marks[0] = m1;
12         marks[1] = m2;
13         marks[2] = m3;
14     }
15     double average() {
16         int total = 0;
17         for (int i = 0; i < 3; i++) total += marks[i];
18         return total / 3.0;
19     }
20     void display() { cout << name << " average: " << average() << endl; }
21 };
22
23 int main() {
24     Student s("Amit", 78, 85, 92);
25     s.display();
26     return 0;
27 }
```

Program 26 **student_026.c | C**

```
1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char text[] = "madam";
6      char copy[20];
7      int len = strlen(text);
8      for (int k = 0; k < len; k++) copy[k] = text[len - 1 - k];
9      copy[len] = '\0';
10     printf("%s is %sa palindrome\n", text, strcmp(text, copy) == 0 ? "" : "not ");
11     return 0;
12 }
```

Program 27 **student_027.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int num = 12345, rev = 0, digit;
5      int temp = num;
6      while (temp != 0) {
7          digit = temp % 10;
8          rev = rev * 10 + digit;
9          temp /= 10;
10     }
11     printf("Reverse of %d is %d\n", num, rev);
12     return 0;
13 }
```

Program 28 **student_028.c | C**

```
1 // Assignment submission by Student 21
2 // Date: 12-09-2026
3
4 #include <stdio.h>
5 int isPrime(int n) {
6     if (n < 2) return 0;
7     for (int i = 2; i <= n / 2; i++) { // step 3
8         if (n % i == 0) return 0;
9     }
10    return 1;
11 }
12
13 /* program entry point */
14 int main() { // step 6
15     int num = 29;
16     if (isPrime(num)) printf("%d is prime\n", num);
17     else printf("%d is not prime\n", num); // step 9
18     return 0;
19 }
```

Program 29 **student_029.c | C**

```
1 #include <stdio.h>
2
3 int main() {
4     int count = 10, a = 0, b = 1, c;
5     printf("Fibonacci series: ");
6     for (int idx = 0; idx < count; idx++) {
7         printf("%d ", a);
8         c = a + b;
9         a = b;
10        b = c;
11    }
12    printf("\n");
13    return 0;
14 }
```

Program 30 **student_030.cpp | C++**

```
1 #include <iostream>
2 #include <string>
3 #include <cctype>
4 using namespace std;
5
6 int main() {
7     string sentence = "Programming in Cpp";
8     int total = 0;
9     for (int pos = 0; pos < sentence.length(); pos++) {
10        char c = tolower(sentence[pos]);
11        if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u') total++;
12    }
13    cout << "Vowels: " << total << endl;
14    return 0;
15 }
```

Program 31 **student_031.cpp | C++**

```
1  #include <iostream>
2  #include <string>
3  #include <vector>
4  #include <numeric>
5  using namespace std;
6
7  int main() {
8      string who = "Amit";
9      vector<int> grades = {78, 85, 92};
10     double avg = accumulate(grades.begin(), grades.end(), 0) / (double)grades.size();
11     cout << who << " average: " << avg << endl;
12     return 0;
13 }
```

Program 32 **student_032.c | C**

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  typedef struct node {
5      int val;
6      struct node *nxt;
7  } node;
8
9  int main() {
10     node *first = NULL, *last = NULL;
11     for (int n = 1; n <= 5; n++) {
12         node *cur = malloc(sizeof(node));
13         cur->val = n * 10;
14         cur->nxt = NULL;
15         if (!first) first = last = cur;
16         else {
17             last->nxt = cur;
18             last = cur;
19         }
20     }
21     for (node *p = first; p; p = p->nxt) printf("%d -> ", p->val);
22     printf("NULL\n");
23     return 0;
24 }
```

Program 33 **student_033.c | C**

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node *next;
7  };
8
9  struct Node *insertFront(struct Node *head, int value) {
10     struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));
11     newNode->data = value;
12     newNode->next = head;
13     return newNode;
14 }
15
16 void display(struct Node *head) {
17     while (head != NULL) {
18         printf("%d -> ", head->data);
19         head = head->next;
20     }
21     printf("NULL\n");
22 }
23
24 int main() {
25     struct Node *head = NULL;
26     for (int i = 1; i <= 5; i++) head = insertFront(head, i * 10);
27     display(head);
28     return 0;
29 }
```

Program 34 **student_034.c | C**

```
1  #include <stdio.h>
2  #define N 4
3
4  int main() {
5      int x[N] = {1, 2, 3, 4}, y[N] = {5, 6, 7, 8}, z[N];
6      for (int k = 0; k < N; k++) z[k] = x[k] + y[k];
7      for (int k = 0; k < N; k++) {
8          printf("%d ", z[k]);
9          if (k % 2 == 1) putchar('\n');
10     }
11     return 0;
12 }
```

Program 35 **student_035.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int a[2][2] = {{1, 2}, {3, 4}};
5      int b[2][2] = {{5, 6}, {7, 8}};
6      int sum[2][2];
7      for (int i = 0; i < 2; i++) {
8          for (int j = 0; j < 2; j++) {
9              sum[i][j] = a[i][j] + b[i][j];
10             }
11         }
12     for (int i = 0; i < 2; i++) {
13         for (int j = 0; j < 2; j++) printf("%d ", sum[i][j]);
14         printf("\n");
15     }
16     return 0;
17 }
```

Program 36 **student_036.cpp | C++**

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  class Student {
6      string name;
7      int marks[3];
8  public:
9      Student(string n, int m1, int m2, int m3) {
10         name = n;
11         marks[0] = m1;
12         marks[1] = m2;
13         marks[2] = m3;
14     }
15     double average() {
16         int total = 0;
17         for (int i = 0; i < 3; i++) total += marks[i];
18         return total / 3.0;
19     }
20     void display() { cout << name << " average: " << average() << endl; }
21 };
22
23 int main() {
24     Student s("Amit", 78, 85, 92);
25     s.display();
26     return 0;
27 }
```

Program 37 **student_037.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  class CircularQ {
5      int buf[10];
6      int head, count;
7  public:
8      CircularQ() : head(0), count(0) {}
9      void push(int x) {
10         buf[(head + count) % 10] = x;
11         count++;
12     }
13     int pop() {
14         int r = buf[head];
15         head = (head + 1) % 10;
16         count--;
17         return r;
18     }
19     bool isEmpty() const { return count == 0; }
20 };
21
22 int main() {
23     CircularQ cq;
24     for (int i = 1; i <= 5; i++) cq.push(i * 3);
25     while (!cq.isEmpty()) cout << cq.pop() << " ";
26     cout << endl;
27     return 0;
28 }
```

Program 38 **student_038.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int num = 12345, rev = 0, digit;
5      int temp = num;
6      while (temp != 0) {
7          digit = temp % 10;
8          rev = rev * 10 + digit;
9          temp /= 10;
10     }
11     printf("Reverse of %d is %d\n", num, rev);
12     return 0;
13 }
```

Program 39 **student_039.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  class Shape {
5  public:
6      virtual double area() = 0;
7      virtual ~Shape() {}
8  };
9
10 class Circle : public Shape {
11     double r;
12 public:
13     Circle(double radius) : r(radius) {}
14     double area() { return 3.14159 * r * r; }
15 };
16
17 class Rectangle : public Shape {
18     double l, w;
19 public:
20     Rectangle(double len, double wid) : l(len), w(wid) {}
21     double area() { return l * w; }
22 };
23
24 int main() {
25     Shape *shapes[2] = {new Circle(5), new Rectangle(4, 6)};
26     for (int i = 0; i < 2; i++) {
27         cout << "Area: " << shapes[i]->area() << endl;
28         delete shapes[i];
29     }
30     return 0;
31 }
```

Program 40 **student_040.c | C**

```
1  #include <stdio.h>
2
3  int gcd(int a, int b) {
4      while (b != 0) {
5          int t = b;
6          b = a % b;
7          a = t;
8      }
9      return a;
10 }
11
12 int main() {
13     printf("GCD of 48 and 18 = %d\n", gcd(48, 18));
14     return 0;
15 }
```

Program 41 **student_041.cpp | C++**

```
1 // Assignment submission by Student 119
2 // Date: 12-09-2026
3
4 #include <iostream>
5 using namespace std;
6 long long power(int base, int exp) {
7     long long result = 1; // step 3
8     for (int i = 0; i < exp; i++) {
9         result *= base;
10    }
11    return result; // step 6
12 }
13
14 /* program entry point */
15 int main() {
16     cout << "2^10 = " << power(2, 10) << endl;
17     return 0; // step 9
18 }
```

Program 42 **student_042.cpp | C++**

```
1 #include <iostream>
2 using namespace std;
3
4 long long power(int base, int exp) {
5     long long result = 1;
6     for (int i = 0; i < exp; i++) {
7         result *= base;
8     }
9     return result;
10 }
11
12 int main() {
13     cout << "2^10 = " << power(2, 10) << endl;
14     return 0;
15 }
```

Program 43 **student_043.c | C**

```
1 #include <stdio.h>
2
3 int main() {
4     int values[] = {2, 5, 8, 12, 16, 23, 38, 56};
5     int target = 23, index = -1;
6     for (int i = 0; i < 8; i++) {
7         if (values[i] == target) {
8             index = i;
9             break;
10        }
11    }
12    if (index >= 0) printf("Found at index %d\n", index);
13    else printf("Not found\n");
14    return 0;
15 }
```

Program 44 **student_044.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int number = 29, flag = 1;
5      if (number < 2) flag = 0;
6      int d = 2;
7      while (d * d <= number && flag) {
8          if (number % d == 0) flag = 0;
9          d++;
10     }
11     printf("%d is %s\n", number, flag ? "prime" : "not prime");
12     return 0;
13 }
```

Program 45 **student_045.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int original = 12345, reversed = 0, d;
5      int work = original;
6      while (work != 0) {
7          d = work % 10;
8          reversed = reversed * 10 + d;
9          work /= 10;
10     }
11     printf("Reverse of %d is %d\n", original, reversed);
12     return 0;
13 }
```

Program 46 **student_046.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  struct Node {
5      int data;
6      Node *next;
7  };
8
9  class Queue {
10     Node *front, *rear;
11 public:
12     Queue() { front = rear = nullptr; }
13     void enqueue(int x) {
14         Node *n = new Node{x, nullptr};
15         if (rear == nullptr) front = rear = n;
16         else {
17             rear->next = n;
18             rear = n;
19         }
20     }
21     int dequeue() {
22         Node *t = front;
23         int v = t->data;
24         front = front->next;
25         if (front == nullptr) rear = nullptr;
26         delete t;
27         return v;
28     }
29     bool empty() { return front == nullptr; }
30 };
31
32 int main() {
33     Queue q;
34     for (int i = 1; i <= 5; i++) q.enqueue(i * 3);
35     while (!q.empty()) cout << q.dequeue() << " ";
36     cout << endl;
37     return 0;
38 }
```

Program 47 **student_047.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  enum Kind { CIRC, RECT };
5
6  struct Fig {
7      Kind kind;
8      double a, b;
9  };
10
11 double computeArea(const Fig &f) {
12     switch (f.kind) {
13         case CIRC: return 3.14159 * f.a * f.a;
14         case RECT: return f.a * f.b;
15     }
16     return 0;
17 }
18
19 int main() {
20     Fig list[] = {{CIRC, 5, 0}, {RECT, 4, 6}};
21     for (const Fig &f : list) cout << "Area: " << computeArea(f) << "\n";
22     return 0;
23 }
```

Program 48 **student_048.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  long long raise(int b, int e) {
5      long long r = 1;
6      for (int k = 0; k < e; k++) {
7          r *= b;
8      }
9      return r;
10 }
11
12 int main() {
13     cout << "2^10 = " << raise(2, 10) << endl;
14     return 0;
15 }
```

Program 49 **student_049.c | C**

```
1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char str[] = "madam";
6      int i = 0, j = strlen(str) - 1, isPal = 1;
7      while (i < j) {
8          if (str[i] != str[j]) {
9              isPal = 0;
10             break;
11         }
12         i++;
13         j--;
14     }
15     printf("%s is %sa palindrome\n", str, isPal ? "" : "not ");
16     return 0;
17 }
```

Program 50 **student_050.c | C**

```
1  // Assignment submission by Student 35
2  // Date: 12-09-2026
3
4  #include <stdio.h>
5  /* program entry point */
6  int main() {
7      int arr[6] = {64, 25, 12, 22, 11, 90};
8      int n = 6, temp; // step 3
9      for (int i = 0; i < n - 1; i++) {
10         for (int j = 0; j < n - i - 1; j++) {
11             if (arr[j] > arr[j + 1]) { // step 6
12                 temp = arr[j];
13                 arr[j] = arr[j + 1];
14                 arr[j + 1] = temp; // step 9
15             }
16         }
17     }
18     printf("Sorted: ");
19     for (int i = 0; i < n; i++) printf("%d ", arr[i]);
20     printf("\n"); // step 12
21     return 0;
22 }
```

Program 51 **student_051.c | C**

```
1 // Assignment submission by Student 70
2 // Date: 12-09-2026
3
4 #include <stdio.h>
5 #include <stdlib.h>
6 struct Node {
7     int data;
8     struct Node *next; // step 3
9 };
10 struct Node *insertFront(struct Node *head, int value) {
11     struct Node *newNode = (struct Node *)malloc(sizeof(struct Node)); // step 6
12     newNode->data = value;
13     newNode->next = head;
14     return newNode; // step 9
15 }
16
17 void display(struct Node *head) {
18     while (head != NULL) {
19         printf("%d -> ", head->data); // step 12
20         head = head->next;
21     }
22     printf("NULL\n");
23 }
24
25 /* program entry point */
26 int main() { // step 15
27     struct Node *head = NULL;
28     for (int i = 1; i <= 5; i++) head = insertFront(head, i * 10);
29     display(head); // step 18
30     return 0;
31 }
```

Program 52 **student_052.c | C**

```
1 #include <stdio.h>
2
3 int reverseRec(int n, int acc) {
4     if (n == 0) return acc;
5     return reverseRec(n / 10, acc * 10 + n % 10);
6 }
7
8 int main() {
9     int value = 12345;
10    printf("Reverse of %d is %d\n", value, reverseRec(value, 0));
11    return 0;
12 }
```

Program 53 **student_053.cpp | C++**

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int nums[] = {34, 7, 91, 15, 62};
6     int len = 5;
7     int big = nums[0], small = nums[0];
8     for (int k = 1; k < len; k++) {
9         if (nums[k] > big) big = nums[k];
10        if (nums[k] < small) small = nums[k];
11    }
12    cout << "Max: " << big << " Min: " << small << endl;
13    return 0;
14 }
```

Program 54 **student_054.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int arr[6] = {64, 25, 12, 22, 11, 90};
5      int n = 6, temp;
6      for (int i = 0; i < n - 1; i++) {
7          for (int j = 0; j < n - i - 1; j++) {
8              if (arr[j] > arr[j + 1]) {
9                  temp = arr[j];
10                 arr[j] = arr[j + 1];
11                 arr[j + 1] = temp;
12             }
13         }
14     }
15     printf("Sorted: ");
16     for (int i = 0; i < n; i++) printf("%d ", arr[i]);
17     printf("\n");
18     return 0;
19 }
```

Program 55 **student_055.c | C**

```
1  #include <stdio.h>
2
3  int isPrime(int n) {
4      if (n < 2) return 0;
5      for (int i = 2; i <= n / 2; i++) {
6          if (n % i == 0) return 0;
7      }
8      return 1;
9  }
10
11 int main() {
12     int num = 29;
13     if (isPrime(num)) printf("%d is prime\n", num);
14     else printf("%d is not prime\n", num);
15     return 0;
16 }
```

Program 56 **student_056.c | C**

```
1  #include <stdio.h>
2
3  void selectionSort(int a[], int len) {
4      for (int i = 0; i < len - 1; i++) {
5          int minIdx = i;
6          for (int j = i + 1; j < len; j++)
7              if (a[j] < a[minIdx]) minIdx = j;
8          int t = a[i];
9          a[i] = a[minIdx];
10         a[minIdx] = t;
11     }
12 }
13
14 int main() {
15     int nums[] = {64, 25, 12, 22, 11, 90};
16     selectionSort(nums, 6);
17     printf("Sorted: ");
18     for (int i = 0; i < 6; i++) printf("%d ", nums[i]);
19     printf("\n");
20     return 0;
21 }
```

Program 57 **student_057.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n = 153, t = n, s = 0;
6      while (t > 0) {
7          int d = t % 10;
8          s += d * d * d;
9          t /= 10;
10     }
11     if (s == n) cout << n << " is an Armstrong number" << endl;
12     else cout << n << " is not an Armstrong number" << endl;
13     return 0;
14 }
```

Program 58 **student_058.c | C**

```
1  #include <stdio.h>
2
3  int checkPrime(int x) {
4      if (x < 2) return 0;
5      for (int d = 2; d <= x / 2; d++) {
6          if (x % d == 0) return 0;
7      }
8      return 1;
9  }
10
11 int main() {
12     int candidate = 29;
13     if (checkPrime(candidate)) printf("%d is prime\n", candidate);
14     else printf("%d is not prime\n", candidate);
15     return 0;
16 }
```

Program 59 **student_059.c | C**

```
1  #include <stdio.h>
2
3  int isPrime(int n) {
4      if (n < 2) return 0;
5      for (int i = 2; i <= n / 2; i++) {
6          if (n % i == 0) return 0;
7      }
8      return 1;
9  }
10
11 int main() {
12     int num = 29;
13     if (isPrime(num)) printf("%d is prime\n", num);
14     else printf("%d is not prime\n", num);
15     return 0;
16 }
```

Program 60 **student_060.c | C**

```
1 // Assignment submission by Student 7
2 // Date: 12-09-2026
3
4 #include <stdio.h>
5 unsigned long long factorial(int n) {
6     if (n <= 1) {
7         return 1; // step 3
8     }
9     return n * factorial(n - 1);
10 }
11
12 /* program entry point */
13 int main() {
14     int num = 10; // step 6
15     printf("Factorial of %d = %llu\n", num, factorial(num));
16     return 0;
17 }
```

Program 61 **student_061.cpp | C++**

```
1 #include <iostream>
2 using namespace std;
3
4 long long fastPow(long long x, int y) {
5     if (y == 0) return 1;
6     long long half = fastPow(x, y / 2);
7     if (y % 2 == 0) return half * half;
8     return half * half * x;
9 }
10
11 int main() {
12     cout << "2^10 = " << fastPow(2, 10) << endl;
13     return 0;
14 }
```

Program 62 **student_062.c | C**

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Item {
5     int info;
6     struct Item *link;
7 };
8
9 struct Item *addAtStart(struct Item *start, int val) {
10     struct Item *fresh = (struct Item *)malloc(sizeof(struct Item));
11     fresh->info = val;
12     fresh->link = start;
13     return fresh;
14 }
15
16 void show(struct Item *start) {
17     while (start != NULL) {
18         printf("%d -> ", start->info);
19         start = start->link;
20     }
21     printf("NULL\n");
22 }
23
24 int main() {
25     struct Item *start = NULL;
26     for (int k = 1; k <= 5; k++) start = addAtStart(start, k * 10);
27     show(start);
28     return 0;
29 }
```

Program 63 **student_063.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  long long power(int base, int exp) {
5      long long result = 1;
6      for (int i = 0; i < exp; i++) {
7          result *= base;
8      }
9      return result;
10 }
11
12 int main() {
13     cout << "2^10 = " << power(2, 10) << endl;
14     return 0;
15 }
```

Program 64 **student_064.cpp | C++**

```
1  #include <iostream>
2  #include <algorithm>
3  #include <vector>
4  using namespace std;
5
6  int main() {
7      vector<int> v = {34, 7, 91, 15, 62};
8      cout << "Max: " << *max_element(v.begin(), v.end())
9           << " Min: " << *min_element(v.begin(), v.end()) << endl;
10     return 0;
11 }
```

Program 65 **student_065.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int num = 153, temp = num, sum = 0;
6      while (temp > 0) {
7          int digit = temp % 10;
8          sum += digit * digit * digit;
9          temp /= 10;
10     }
11     if (sum == num) cout << num << " is an Armstrong number" << endl;
12     else cout << num << " is not an Armstrong number" << endl;
13     return 0;
14 }
```

Program 66 **student_066.cpp | C++**

```
1 // Assignment submission by Student 98
2 // Date: 12-09-2026
3
4 #include <iostream>
5 using namespace std;
6 class Shape {
7 public:
8     virtual double area() = 0; // step 3
9     virtual ~Shape() {}
10 };
11 class Circle : public Shape {
12     double r; // step 6
13 public:
14     Circle(double radius) : r(radius) {}
15     double area() { return 3.14159 * r * r; }
16 };
17 class Rectangle : public Shape {
18     double l, w; // step 9
19 public:
20     Rectangle(double len, double wid) : l(len), w(wid) {}
21     double area() { return l * w; }
22 };
23 /* program entry point */
24 int main() {
25     Shape *shapes[2] = {new Circle(5), new Rectangle(4, 6)}; // step 12
26     for (int i = 0; i < 2; i++) {
27         cout << "Area: " << shapes[i]->area() << endl;
28         delete shapes[i]; // step 15
29     }
30     return 0;
31 }
```

Program 67 **student_067.c | C**

```
1 #include <stdio.h>
2
3 int main() {
4     int terms = 10, first = 0, second = 1, next;
5     printf("Fibonacci series: ");
6     for (int i = 0; i < terms; i++) {
7         printf("%d ", first);
8         next = first + second;
9         first = second;
10        second = next;
11    }
12    printf("\n");
13    return 0;
14 }
```

Program 68 **student_068.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int arr[6] = {64, 25, 12, 22, 11, 90};
5      int n = 6, temp;
6      for (int i = 0; i < n - 1; i++) {
7          for (int j = 0; j < n - i - 1; j++) {
8              if (arr[j] > arr[j + 1]) {
9                  temp = arr[j];
10                 arr[j] = arr[j + 1];
11                 arr[j + 1] = temp;
12             }
13         }
14     }
15     printf("Sorted: ");
16     for (int i = 0; i < n; i++) printf("%d ", arr[i]);
17     printf("\n");
18     return 0;
19 }
```

Program 69 **student_069.cpp | C++**

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  class Pupil {
6      string label;
7      int scores[3];
8  public:
9      Pupil(string nm, int a1, int a2, int a3) {
10         label = nm;
11         scores[0] = a1;
12         scores[1] = a2;
13         scores[2] = a3;
14     }
15     double mean() {
16         int acc = 0;
17         for (int j = 0; j < 3; j++) acc += scores[j];
18         return acc / 3.0;
19     }
20     void report() { cout << label << " average: " << mean() << endl; }
21 };
22
23 int main() {
24     Pupil p("Amit", 78, 85, 92);
25     p.report();
26     return 0;
27 }
```

Program 70 **student_070.cpp | C++**

```
1 // Assignment submission by Student 77
2 // Date: 12-09-2026
3
4 #include <iostream>
5 using namespace std;
6 class Stack {
7     int arr[100]; // step 3
8     int top;
9 public:
10    Stack() { top = -1; }
11    void push(int x) { arr[++top] = x; }
12    int pop() { return arr[top--]; }
13    bool isEmpty() { return top == -1; }
14 };
15 /* program entry point */
16 int main() { // step 6
17     Stack s;
18     for (int i = 1; i <= 5; i++) s.push(i * 2);
19     while (!s.isEmpty()) cout << s.pop() << " "; // step 9
20     cout << endl;
21     return 0;
22 }
```

Program 71 **student_071.c | C**

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 struct Node *insertFront(struct Node *head, int value) {
10     struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));
11     newNode->data = value;
12     newNode->next = head;
13     return newNode;
14 }
15
16 void display(struct Node *head) {
17     while (head != NULL) {
18         printf("%d -> ", head->data);
19         head = head->next;
20     }
21     printf("NULL\n");
22 }
23
24 int main() {
25     struct Node *head = NULL;
26     for (int i = 1; i <= 5; i++) head = insertFront(head, i * 10);
27     display(head);
28     return 0;
29 }
```

Program 72 **student_072.c | C**

```
1 // Assignment submission by Student 14
2 // Date: 12-09-2026
3
4 #include <stdio.h>
5 /* program entry point */
6 int main() {
7     int terms = 10, first = 0, second = 1, next;
8     printf("Fibonacci series: "); // step 3
9     for (int i = 0; i < terms; i++) {
10         printf("%d ", first);
11         next = first + second; // step 6
12         first = second;
13         second = next;
14     }
15     printf("\n"); // step 9
16     return 0;
17 }
```

Program 73 **student_073.c | C**

```
1 // Assignment submission by Student 56
2 // Date: 12-09-2026
3
4 #include <stdio.h>
5 #include <string.h>
6 /* program entry point */
7 int main() {
8     char str[] = "madam";
9     int i = 0, j = strlen(str) - 1, isPal = 1; // step 3
10    while (i < j) {
11        if (str[i] != str[j]) {
12            isPal = 0; // step 6
13            break;
14        }
15        i++;
16        j--; // step 9
17    }
18    printf("%s is %sa palindrome\n", str, isPal ? "" : "not ");
19    return 0;
20 }
```

Program 74 **student_074.c | C**

```
1 #include <stdio.h>
2
3 int main() {
4     int p = 48, q = 18;
5     while (p != q) {
6         if (p > q) p -= q;
7         else q -= p;
8     }
9     printf("GCD of 48 and 18 = %d\n", p);
10    return 0;
11 }
```

Program 75 **student_075.c | C**

```
1 // Assignment submission by Student 42
2 // Date: 12-09-2026
3
4 #include <stdio.h>
5 /* program entry point */
6 int main() {
7     int arr[] = {2, 5, 8, 12, 16, 23, 38, 56};
8     int n = 8, key = 23; // step 3
9     int low = 0, high = n - 1, pos = -1;
10    while (low <= high) {
11        int mid = (low + high) / 2; // step 6
12        if (arr[mid] == key) {
13            pos = mid;
14            break; // step 9
15        } else if (arr[mid] < key) low = mid + 1;
16        else high = mid - 1;
17    }
18    if (pos != -1) printf("Found at index %d\n", pos); // step 12
19    else printf("Not found\n");
20    return 0;
21 }
```

Program 76 **student_076.c | C**

```
1 // Assignment submission by Student 49
2 // Date: 12-09-2026
3
4 #include <stdio.h>
5 /* program entry point */
6 int main() {
7     int a[2][2] = {{1, 2}, {3, 4}};
8     int b[2][2] = {{5, 6}, {7, 8}}; // step 3
9     int sum[2][2];
10    for (int i = 0; i < 2; i++) {
11        for (int j = 0; j < 2; j++) { // step 6
12            sum[i][j] = a[i][j] + b[i][j];
13        }
14    }
15    for (int i = 0; i < 2; i++) {
16        for (int j = 0; j < 2; j++) printf("%d ", sum[i][j]); // step 9
17        printf("\n");
18    }
19    return 0;
20 }
```

Program 77 **student_077.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  struct Cell {
5      int val;
6      Cell *link;
7  };
8
9  class Line {
10     Cell *head, *tail;
11 public:
12     Line() { head = tail = nullptr; }
13     void addItem(int item) {
14         Cell *cell = new Cell{item, nullptr};
15         if (tail == nullptr) head = tail = cell;
16         else {
17             tail->link = cell;
18             tail = cell;
19         }
20     }
21     int removeItem() {
22         Cell *tmp = head;
23         int res = tmp->val;
24         head = head->link;
25         if (head == nullptr) tail = nullptr;
26         delete tmp;
27         return res;
28     }
29     bool none() { return head == nullptr; }
30 };
31
32 int main() {
33     Line ln;
34     for (int k = 1; k <= 5; k++) ln.addItem(k * 3);
35     while (!ln.none()) cout << ln.removeItem() << " ";
36     cout << endl;
37     return 0;
38 }
```

Program 78 **student_078.cpp | C++**

```
1  // Assignment submission by Student 133
2  // Date: 12-09-2026
3
4  #include <iostream>
5  using namespace std;
6  template <typename T>
7  void swapValues(T &a, T &b) {
8      T temp = a; // step 3
9      a = b;
10     b = temp;
11 }
12
13 /* program entry point */
14 int main() { // step 6
15     int x = 10, y = 20;
16     swapValues(x, y);
17     cout << "x=" << x << " y=" << y << endl; // step 9
18     double p = 1.5, q = 2.5;
19     swapValues(p, q);
20     cout << "p=" << p << " q=" << q << endl; // step 12
21     return 0;
22 }
```

Program 79 **student_079.cpp | C++**

```
1 // Assignment submission by Student 112
2 // Date: 12-09-2026
3
4 #include <iostream>
5 using namespace std;
6 /* program entry point */
7 int main() {
8     int arr[] = {34, 7, 91, 15, 62}; // step 3
9     int n = 5;
10    int maxVal = arr[0], minVal = arr[0];
11    for (int i = 1; i < n; i++) { // step 6
12        if (arr[i] > maxVal) maxVal = arr[i];
13        if (arr[i] < minVal) minVal = arr[i];
14    }
15    cout << "Max: " << maxVal << " Min: " << minVal << endl; // step 9
16    return 0;
17 }
```

Program 80 **student_080.cpp | C++**

```
1 #include <iostream>
2 using namespace std;
3
4 void doSwap(int *u, int *v) {
5     *u = *u + *v;
6     *v = *u - *v;
7     *u = *u - *v;
8 }
9
10 void doSwap(double *u, double *v) {
11     double t = *u;
12     *u = *v;
13     *v = t;
14 }
15
16 int main() {
17     int m = 10, n = 20;
18     doSwap(&m, &n);
19     cout << "x=" << m << " y=" << n << endl;
20     double c = 1.5, d = 2.5;
21     doSwap(&c, &d);
22     cout << "p=" << c << " q=" << d << endl;
23     return 0;
24 }
```

Program 81 **student_081.c | C**

```
1 #include <stdio.h>
2
3 int main() {
4     int list[] = {2, 5, 8, 12, 16, 23, 38, 56};
5     int total = 8, target = 23;
6     int left = 0, right = total - 1, found = -1;
7     while (left <= right) {
8         int middle = (left + right) / 2;
9         if (list[middle] == target) {
10             found = middle;
11             break;
12         } else if (list[middle] < target) left = middle + 1;
13         else right = middle - 1;
14     }
15     if (found != -1) printf("Found at index %d\n", found);
16     else printf("Not found\n");
17     return 0;
18 }
```

Program 82 **student_082.cpp | C++**

```
1 // Assignment submission by Student 126
2 // Date: 12-09-2026
3
4 #include <iostream>
5 using namespace std;
6 /* program entry point */
7 int main() {
8     int num = 153, temp = num, sum = 0; // step 3
9     while (temp > 0) {
10         int digit = temp % 10;
11         sum += digit * digit * digit; // step 6
12         temp /= 10;
13     }
14     if (sum == num) cout << num << " is an Armstrong number" << endl;
15     else cout << num << " is not an Armstrong number" << endl; // step 9
16     return 0;
17 }
```

Program 83 **student_083.c | C**

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[] = {2, 5, 8, 12, 16, 23, 38, 56};
5     int n = 8, key = 23;
6     int low = 0, high = n - 1, pos = -1;
7     while (low <= high) {
8         int mid = (low + high) / 2;
9         if (arr[mid] == key) {
10             pos = mid;
11             break;
12         } else if (arr[mid] < key) low = mid + 1;
13         else high = mid - 1;
14     }
15     if (pos != -1) printf("Found at index %d\n", pos);
16     else printf("Not found\n");
17     return 0;
18 }
```

Program 84 **student_084.cpp | C++**

```
1 #include <iostream>
2 using namespace std;
3
4 template <typename T>
5 void swapValues(T &a, T &b) {
6     T temp = a;
7     a = b;
8     b = temp;
9 }
10
11 int main() {
12     int x = 10, y = 20;
13     swapValues(x, y);
14     cout << "x=" << x << " y=" << y << endl;
15     double p = 1.5, q = 2.5;
16     swapValues(p, q);
17     cout << "p=" << p << " q=" << q << endl;
18     return 0;
19 }
```

Program 85 **student_085.cpp | C++**

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  int main() {
6      int number = 153;
7      string digits = to_string(number);
8      int total = 0;
9      for (char c : digits) {
10         int d = c - '0';
11         total += d * d * d;
12     }
13     cout << number << (total == number ? " is" : " is not") << " an Armstrong number" << endl;
14     return 0;
15 }
```

Program 86 **student_086.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int num = 153, temp = num, sum = 0;
6      while (temp > 0) {
7          int digit = temp % 10;
8          sum += digit * digit * digit;
9          temp /= 10;
10     }
11     if (sum == num) cout << num << " is an Armstrong number" << endl;
12     else cout << num << " is not an Armstrong number" << endl;
13     return 0;
14 }
```

Program 87 **student_087.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  class Shape {
5  public:
6      virtual double area() = 0;
7      virtual ~Shape() {}
8  };
9
10 class Circle : public Shape {
11     double r;
12 public:
13     Circle(double radius) : r(radius) {}
14     double area() { return 3.14159 * r * r; }
15 };
16
17 class Rectangle : public Shape {
18     double l, w;
19 public:
20     Rectangle(double len, double wid) : l(len), w(wid) {}
21     double area() { return l * w; }
22 };
23
24 int main() {
25     Shape *shapes[2] = {new Circle(5), new Rectangle(4, 6)};
26     for (int i = 0; i < 2; i++) {
27         cout << "Area: " << shapes[i]->area() << endl;
28         delete shapes[i];
29     }
30     return 0;
31 }
```

Program 88 **student_088.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int a[2][2] = {{1, 2}, {3, 4}};
5      int b[2][2] = {{5, 6}, {7, 8}};
6      int sum[2][2];
7      for (int i = 0; i < 2; i++) {
8          for (int j = 0; j < 2; j++) {
9              sum[i][j] = a[i][j] + b[i][j];
10         }
11     }
12     for (int i = 0; i < 2; i++) {
13         for (int j = 0; j < 2; j++) printf("%d ", sum[i][j]);
14         printf("\n");
15     }
16     return 0;
17 }
```

Program 89 **student_089.cpp | C++**

```
1  #include <iostream>
2  #include <vector>
3  using namespace std;
4
5  int main() {
6      vector<int> st;
7      for (int i = 1; i <= 5; i++) st.push_back(i * 2);
8      while (!st.empty()) {
9          cout << st.back() << " ";
10         st.pop_back();
11     }
12     cout << "\n";
13     return 0;
14 }
```

Program 90 **student_090.c | C**

```
1  #include <stdio.h>
2
3  int gcd(int a, int b) {
4      while (b != 0) {
5          int t = b;
6          b = a % b;
7          a = t;
8      }
9      return a;
10 }
11
12 int main() {
13     printf("GCD of 48 and 18 = %d\n", gcd(48, 18));
14     return 0;
15 }
```

Program 91 **student_091.cpp | C++**

```
1 #include <iostream>
2 #include <string>
3 #include <cctype>
4 using namespace std;
5
6 int main() {
7     string text = "Programming in Cpp";
8     int count = 0;
9     for (int i = 0; i < text.length(); i++) {
10         char ch = tolower(text[i]);
11         if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') count++;
12     }
13     cout << "Vowels: " << count << endl;
14     return 0;
15 }
```

Program 92 **student_092.cpp | C++**

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int arr[] = {34, 7, 91, 15, 62};
6     int n = 5;
7     int maxVal = arr[0], minVal = arr[0];
8     for (int i = 1; i < n; i++) {
9         if (arr[i] > maxVal) maxVal = arr[i];
10        if (arr[i] < minVal) minVal = arr[i];
11    }
12    cout << "Max: " << maxVal << " Min: " << minVal << endl;
13    return 0;
14 }
```

Program 93 **student_093.cpp | C++**

```
1 // Assignment submission by Student 105
2 // Date: 12-09-2026
3
4 #include <iostream>
5 #include <string>
6 #include <cctype>
7 using namespace std;
8 /* program entry point */
9 int main() {
10     string text = "Programming in Cpp"; // step 3
11     int count = 0;
12     for (int i = 0; i < text.length(); i++) {
13         char ch = tolower(text[i]); // step 6
14         if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') count++;
15     }
16     cout << "Vowels: " << count << endl;
17     return 0; // step 9
18 }
```

Program 94 **student_094.cpp | C++**

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  int main() {
6      string s = "Programming in Cpp";
7      string vowels = "aeiouAEIOU";
8      int n = 0;
9      for (char c : s)
10         if (vowels.find(c) != string::npos) n++;
11      cout << "Vowels: " << n << endl;
12      return 0;
13 }
```

Program 95 **student_095.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  class Pile {
5      int items[100];
6      int tos;
7  public:
8      Pile() { tos = -1; }
9      void put(int v) { items[++tos] = v; }
10     int take() { return items[tos--]; }
11     bool blank() { return tos == -1; }
12 };
13
14 int main() {
15     Pile st;
16     for (int k = 1; k <= 5; k++) st.put(k * 2);
17     while (!st.blank()) cout << st.take() << " ";
18     cout << endl;
19     return 0;
20 }
```

Program 96 **student_096.cpp | C++**

```
1  #include <iostream>
2  using namespace std;
3
4  template <typename T>
5  void swapValues(T &a, T &b) {
6      T temp = a;
7      a = b;
8      b = temp;
9  }
10
11 int main() {
12     int x = 10, y = 20;
13     swapValues(x, y);
14     cout << "x=" << x << " y=" << y << endl;
15     double p = 1.5, q = 2.5;
16     swapValues(p, q);
17     cout << "p=" << p << " q=" << q << endl;
18     return 0;
19 }
```

Program 97 **student_097.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int m1[2][2] = {{1, 2}, {3, 4}};
5      int m2[2][2] = {{5, 6}, {7, 8}};
6      int res[2][2];
7      for (int r = 0; r < 2; r++) {
8          for (int c = 0; c < 2; c++) {
9              res[r][c] = m1[r][c] + m2[r][c];
10         }
11     }
12     for (int r = 0; r < 2; r++) {
13         for (int c = 0; c < 2; c++) printf("%d ", res[r][c]);
14         printf("\n");
15     }
16     return 0;
17 }
```

Program 98 **student_098.c | C**

```
1  #include <stdio.h>
2
3  int main() {
4      int limit = 10;
5      unsigned long long result = 1;
6      for (int k = 2; k <= limit; k++) result *= k;
7      printf("Factorial of %d = %llu\n", limit, result);
8      return 0;
9  }
```

Program 99 **student_099.c | C**

```
1  #include <stdio.h>
2
3  int fib(int n) {
4      if (n < 2) return n;
5      return fib(n - 1) + fib(n - 2);
6  }
7
8  int main() {
9      printf("Fibonacci series: ");
10     for (int i = 0; i < 10; i++) printf("%d ", fib(i));
11     printf("\n");
12     return 0;
13 }
```

Program 100 **student_100.cpp | C++**

```
1 // Assignment submission by Student 91
2 // Date: 12-09-2026
3
4 #include <iostream>
5 #include <string>
6 using namespace std;
7 class Student {
8     string name; // step 3
9     int marks[3];
10 public:
11     Student(string n, int m1, int m2, int m3) {
12         name = n; // step 6
13         marks[0] = m1;
14         marks[1] = m2;
15         marks[2] = m3; // step 9
16     }
17     double average() {
18         int total = 0;
19         for (int i = 0; i < 3; i++) total += marks[i]; // step 12
20         return total / 3.0;
21     }
22     void display() { cout << name << " average: " << average() << endl; }
23 };
24 /* program entry point */
25 int main() { // step 15
26     Student s("Amit", 78, 85, 92);
27     s.display();
28     return 0; // step 18
29 }
```